

DSAA3073: Theories in Data Science

Week 3 Validation Sheet: Agnostic PAC Learning, Uniform Convergence & Fundamental Theorem

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Each question is worth 5 points. Select the single best answer.

Question 1. [5 marks] In agnostic PAC learning, what does “agnostic” mean?

- A) The algorithm doesn't know the data distribution.
- B) The algorithm doesn't assume there exists a perfect hypothesis with zero error in $\mathcal{H}$.
- C) The algorithm uses a uniform prior over all hypotheses.
- D) The learning problem has no labels.

Question 2. [5 marks] Which statement correctly describes uniform convergence?

- A) Each hypothesis converges to its true error at a different rate.
- B) The worst-case deviation between empirical and true error over all hypotheses goes to zero as sample size increases.
- C) All hypotheses in $\mathcal{H}$ must have the same empirical error.
- D) Convergence only happens for the hypothesis with minimum empirical risk.

Question 3. [5 marks] According to the Fundamental Theorem of Statistical Learning, which is NOT equivalent to the others?

- A) $\mathcal{H}$ has finite VC dimension.
- B) $\mathcal{H}$ is agnostic PAC-learnable.
- C) ERM algorithm can be implemented in polynomial time.
- D) $\mathcal{H}$ has uniform convergence property.

Question 4. [5 marks] Why do we need uniform convergence for ERM to succeed in agnostic learning?

- A) To ensure the training error is zero.
- B) To guarantee that empirical risk reliably estimates true risk for all hypotheses simultaneously, including the ERM choice.
- C) To make the algorithm run faster.
- D) To reduce the VC dimension of the hypothesis class.

Question 5. [5 marks] The sample complexity for agnostic PAC learning with VC dimension d is:

- A) $O\left(\frac{\log d}{\epsilon^2}\right)$
- B) $O\left(\frac{d \log d}{\epsilon}\right)$
- C) $O\left(\frac{d}{\epsilon^2}\right)$
- D) $O\left(\frac{d^2}{\epsilon}\right)$

Question 6. [5 marks] What does the No Free Lunch Theorem tell us?

- A) Free algorithms don't work as well as paid ones.
- B) If $\text{VCdim}(\mathcal{H}) = \infty$, then no algorithm can PAC-learn $\mathcal{H}$.
- C) All hypothesis classes with finite VC dimension have the same sample complexity.
- D) Learning requires labeled data.

Question 7. [5 marks] In the symmetrization proof technique, we introduce a ghost sample to:

- A) Double the effective sample size.
- B) Replace the unknown expectation $R(h)$ with an empirical average on an independent sample.
- C) Satisfy the independence assumptions of Hoeffding's inequality.
- D) Implement the ERM algorithm more efficiently.

Question 8. [5 marks] Which hypothesis class is NOT PAC-learnable?

- A) Linear classifiers in $\mathbb{R}^{\{1000\}}$
- B) Polynomial classifiers of degree 5 in $\mathbb{R}^{\{10\}}$
- C) Axis-aligned rectangles in $\mathbb{R}^{\{50\}}$
- D) $\mathcal{H} = \{\text{sign}(\sin(\omega x)) : \omega \in \mathbb{R}_+\}$ on $\mathbb{R}$

Part B: Mixed Format Questions (90 points)

Question 9. [12 marks] Define and compare realizable vs agnostic PAC learning.

- (a) [4 marks] State the realizability assumption and what it means practically.
- (b) [4 marks] Compare the learning goals: what does each setting try to achieve?
- (c) [4 marks] Compare sample complexities: for a class with $|\mathcal{H}| = 5000$, $\varepsilon = 0.1$, $\delta = 0.05$, compute sample size for both settings.

Question 10. [15 marks] Prove that uniform convergence implies ERM succeeds.

- (a) [5 marks] State precisely what it means for “ERM to succeed” in the agnostic setting.
- (b) [10 marks] Assume uniform convergence holds with tolerance $\frac{\varepsilon}{2}$: $\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \frac{\varepsilon}{2}$. Prove that ERM returns h_S with $R(h_S) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$.

Question 11. [13 marks] Explain the four equivalent conditions in the Fundamental Theorem of Statistical Learning.

- (a) [5 marks] List the four conditions and state that they are equivalent.
- (b) [4 marks] Explain the intuition for why “finite VC dimension $\rightarrow$ uniform convergence.”
- (c) [4 marks] Explain the intuition for why “realizable PAC learnable $\rightarrow$ finite VC dimension” (No Free Lunch direction).

Question 12. [15 marks] Derive the agnostic PAC sample complexity bound from uniform convergence.

(a) [8 marks] Starting from Hoeffding's inequality for a single hypothesis, derive the uniform convergence bound for finite $\mathcal{H}$.

(b) [7 marks] For $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 15$, using the VC-based bound $m = O\left(\frac{d}{\varepsilon^2}\right)$ (ignoring log factors), compute the required sample size for $\varepsilon = 0.1$. How does this compare to a finite class with $|\mathcal{H}| = 2^{\{15\}}$?

Question 13. [10 marks] Explain the symmetrization technique.

- (a) [4 marks] What is the main idea of symmetrization and why is it useful?
- (b) [6 marks] In the symmetrization lemma, we bound $\mathbb{E}\left[\sup_{h \in \mathcal{H}} (R(h) - \hat{R}_S(h))\right]$ using a ghost sample. Explain the key steps that allow us to go from this expectation to a bound involving $\Pi_{\mathcal{H}}(2m)$.

Question 14. [15 marks] The No Free Lunch Theorem and its implications.

- (a) [6 marks] State the No Free Lunch Theorem and explain what it means for the possibility of universal learning algorithms.
- (b) [5 marks] Sketch the adversarial construction: if $\text{VCdim}(\mathcal{H}) = \infty$, how does the adversary construct a distribution that defeats any learning algorithm?
- (c) [4 marks] Does No Free Lunch contradict the universality of neural networks? Explain.

Question 15. [10 marks] Application problem: model selection.

- (a) [5 marks] You have three hypothesis classes: $\mathcal{H}_1$ with $\text{VCdim} = 5$, $\mathcal{H}_2$ with $\text{VCdim} = 20$, $\mathcal{H}_3$ with $\text{VCdim} = 100$. You have $m = 1000$ samples. For $\varepsilon = 0.1$, which classes are “learnable” with this sample size (i.e., sample complexity is satisfied)?
- (b) [5 marks] Explain the trade-off: why not always choose the simplest class ($\mathcal{H}_1$)? Why not always choose the most expressive class ($\mathcal{H}_3$)?

— End of Validation Sheet —